

## 2601-MTL106/MTL2006: TUTORIAL-2

**Q1.** A fair die is rolled twice. Let  $X$  denote the sum of the face values obtained on the two rolls, and let  $Y$  denote the absolute difference between the face values. Describe the sample space  $\Omega$ . Determine whether  $X$  and  $Y$  are random variables on  $\Omega$ . Justify your answer.

**Q2.** Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space, and let  $\mathbb{X}$  denote the set of all real-valued random variables defined on this probability space.

- Show that  $\mathbb{X}$  is a vector space over  $\mathbb{R}$ .
- Assume that the sample space  $\Omega$  is finite. Determine a basis for the vector space  $\mathbb{X}$  and hence find its dimension.

**Q3.** Determine whether each of the following functions is a cumulative distribution function (distribution function). Justify your answer in each case.

$$\begin{aligned} \text{a) } F(x) &= \begin{cases} 0, & x < 0, \\ x, & 0 \leq x \leq \frac{1}{2}, \\ 1, & x > \frac{1}{2}. \end{cases} & \text{b) } F(x) &= \frac{1}{\pi} \tan^{-1}(x), \quad -\infty < x < \infty, \\ \text{c) } F(x) &= \begin{cases} 0, & -\infty < x < 0, \\ 1 - e^{-x}, & 0 \leq x < \infty. \end{cases} \end{aligned}$$

**Q4.** Let  $F$  and  $G$  be cumulative distribution functions (CDFs).

- Show that  $H(x) = \lambda F(x) + (1 - \lambda)G(x)$ ,  $0 \leq \lambda \leq 1$ , is also a CDF.
- Determine whether the product  $H(x) = F(x)G(x)$  is necessarily a CDF. Justify your answer.
- Determine whether the function  $H(x) = F(x) + (1 - F(x)) \log(1 - F(x))$  is necessarily a CDF. Justify your answer.

**Q5.** Let  $f$  and  $g$  be probability density functions (PDFs).

- Show that  $h(x) = \lambda f(x) + (1 - \lambda)g(x)$ ,  $0 \leq \lambda \leq 1$ , is also a density function.
- Determine whether the product  $h(x) = f(x)g(x)$  is necessarily a probability density function. Justify your answer.

**Q6.** Let  $X$  be a continuous type random variable with PDF

$$f(x) = \begin{cases} \alpha + \beta x^2, & 0 \leq x \leq 2 \\ 0, & \text{otherwise,} \end{cases}$$

where  $\alpha$  and  $\beta$  are constants. If  $\mathbb{E}[X] = \frac{4}{3}$ , determine the values of  $\alpha$  and  $\beta$ .

**Q7.** Let  $X$  be a non-negative random variable with CDF

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{x}{2\pi}, & 0 \leq x < 2\pi, \\ 1, & x \geq 2\pi. \end{cases}$$

Find the CDF of  $Y := X^2$ . Determine whether  $X$  and  $Y$  are continuous random variables. Justify your answer.

**Q8.** Let  $X$  be a random variable with CDF

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{x}{2}, & 0 \leq x < 1, \\ \frac{2}{3}, & 1 \leq x < 2, \\ \frac{11}{12}, & 2 \leq x < 3, \\ 1, & x \geq 3. \end{cases}$$

Using the given CDF, compute the following probabilities:

$$i) \mathbb{P}\left(X > \frac{1}{2}\right), \quad ii) \mathbb{P}(2 < X \leq 4), \quad iii) \mathbb{P}(X < 3), \quad iv) \mathbb{P}(X = 1).$$

**Q9.** Let  $X$  be a random variable satisfying  $\mathbb{P}(0 \leq X \leq c) = 1$ , where  $c$  is a constant. Show that

$$\text{Var}(X) \leq \frac{c^2}{4}.$$

**Q10.** Let  $X$  be a positive random variable with PDF  $f_X(x)$ . Define a random variable  $U := \frac{X}{1+X}$ . Determine the PDF of  $U$ .

**Q11.** Let  $X$  be a random variable with pdf

$$f(x) = \begin{cases} \frac{\beta 5^\beta}{x^{\beta+1}}, & x \geq 5, \\ 0, & x < 5, \end{cases}$$

where  $\beta > 0$ . Show that  $\mathbb{E}[X^n]$  exists if and only if  $n < \beta$ . Assuming that  $\beta > 2$ , determine the mean and variance of the random variable  $X$ .

**Q12.** Let  $X$  be a non-negative integer-valued random variable with probability generating function (pgf)  $G_X(\cdot)$ . Define

$$Y := \alpha X + \beta$$

where  $\alpha$  and  $\beta$  are non-negative integers. Determine the probability generating function of  $Y$  in terms of  $G_X(\cdot)$ .

**Q13.** Let  $X$  be a discrete random variable with probability generating function

$$G_X(s) = \frac{s}{3}(2 + s^2).$$

Determine the probability mass function (pmf) of  $X$ .

**Q14.** Let  $X$  be a random variable with probability density function

$$f_X(x) = \begin{cases} e^{-2x} + \frac{1}{2}e^{-x}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

- i) Determine the moment generating function (mgf) of  $X$ .
- ii) Using the mgf, compute the first and second moments of  $X$  i.e., find  $\mathbb{E}[X]$  and  $\mathbb{E}[X^2]$ .

**Q15.** Let  $X$  be a Cauchy random variable with probability density function

$$f(x) = \frac{1}{\pi} \frac{1}{1+x^2}, \quad -\infty < x < \infty.$$

Determine whether the moment generating function (mgf) of  $X$  exists. Justify your answer.

**Q16.** Let  $X$  be a discrete random variable with moment generating function

$$m_X(t) = a + be^{2t} + ce^{4t},$$

where  $a$ ,  $b$ , and  $c$  are constants. Suppose that

$$\mathbb{E}[X] = 3, \quad \text{Var}(X) = 2.$$

Determine  $\mathbb{E}[2^X]$ .

**Q17.** Let  $\phi_X(\cdot)$  be the characteristic function of a random variable  $X$ . Show that, for any  $u \in \mathbb{R}$ ,

$$1 - |\phi_X(2u)|^2 \leq 4 \left( 1 - |\phi_X(u)|^2 \right).$$

**Q18.** Let  $X$  be a continuous random variable with cumulative distribution function  $F_X(\cdot)$ . Define  $Y := F_X(X)$ . Determine the distribution of  $Y$  and its characteristic function.